Catalytic QUBO paths for quantum hypergraph partitioning

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Abstract

Hypergraph partitioning assigns interacting vertices to balanced blocks while minimizing dependencies between them. Classical multilevel methods provide strong solutions. Quantum approaches offer alternative dynamics, but must encode many-vertex costs and enforce balance. Faithful hypergraph encoding alone does not ensure useful balanced search. We construct temporary incidence-dependent QUBO interactions that preserve the cut-net endpoint and control intermediate variational response. A classical free-energy machine tests the construction on 30 original hypergraphs with one shared 48-replica policy. It improves 15 KaHyPar references at exact balance, but none at 1% or 3% imbalance. Small coherent simulations show changed optimal-partition probabilities; circuit dataflow relates cuts to communication. Preprocessing costs and precision sensitivity limit practical gains. These results support catalytic path design without establishing quantum advantage.

Keywords: quantum optimization, QUBO, hypergraph partitioning, catalysts, free-energy machine

Introduction

Hypergraph partitioning allocates interacting tasks or components to balanced blocks while limiting shared resources. A net represents a signal or dependency involving many vertices; the cut-net objective charges it once when it crosses blocks. This model is used for circuit netlists and distributed computations[1,2]. Multilevel methods combine contraction and local refinement[3,4], and KaHyPar and its parallel successor provide strong classical references[5,6]. For an energy-based solver, the challenge is to preserve the many-vertex objective while creating useful motion toward a balanced partition. Encoding and dynamics answer different questions: what is minimized, and which states can be reached.

Quantum annealing searches through time-dependent Hamiltonians[7–9], while quantum approximate optimization algorithms (QAOA) use alternating parameterized circuits[10]. Ising formulations connect these methods to discrete optimization[11]. Quadratic unconstrained binary optimization (QUBO) uses two-body interactions; higher-order unconstrained binary optimization (HUBO) retains many-body terms. Exact quadratization trades interaction order for auxiliary variables[12–14]. Graph partitioning has been tested on D-Wave hardware and through hybrid QUBO decomposition[15], while constraint-preserving mixers and drivers address fixed block sizes[16,17].

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These constructions motivate hypergraph optimization, but an edge-cut result does not establish performance on the many-pin cut-net objective.

Recent hypergraph studies make this distinction explicit. Li et al. compare objective encodings and test simulated QAOA and annealing on small 3-uniform inputs[18]. Ibrahim et al. optimize distributions over partitions for fairness[19]; their target differs from cut-net minimization. These studies leave a search question unresolved: how can a faithful quantum-compatible energy support effective balanced search on real hypergraphs? Preserving the target minimum does not determine the response along the search path. The representation and balance constraints also shape the intermediate dynamics, so endpoint fidelity alone is insufficient.

Temporary catalyst interactions offer a way to address the path while restoring the target energy at the endpoint. In quantum annealing, such interactions can alter transitions or gaps[20–22], with benefits depending on the model and schedule[23,24]. For hypergraphs, incidence structure supplies both local couplings and extended balanced directions. The question is whether these directions can strengthen the intermediate response without replacing the objective. Overlap-gap theory provides a broader motivation: disconnected ranges of near-optimal overlaps can obstruct specified stable algorithms on specified random ensembles[25–27]. Related restrictions apply to shallow local quantum algorithms[28,29], and recent work compares quantum and mean-field-inspired searches[30]. These results do not establish an overlap gap for the real netlists studied here; our mechanism analysis concerns local response rather than a global escape theorem.

We construct temporary incidence-dependent QUBO interactions that preserve the cut-net endpoint while controlling balanced variational response. Exact examples identify objective distortion; a reduced free-energy Hessian connects the catalyst to local stability; and collective modes extend the available directions. A free-energy machine (FEM)[31] supplies a scalable classical testbed. On 30 original hypergraphs, one shared policy improves 15 KaHyPar references at exact balance, but none at 1% or 3% imbalance. The real-instance comparisons assess practical quality and cost, small coherent simulations separately test quantum dynamics, and circuit dataflow gives cut quality a communication interpretation. Preprocessing costs and numerical sensitivity constrain the practical gains; the classical and coherent results do not establish quantum advantage.

Results

Preserving the cut-net objective

Two-way hypergraph partitioning divides a vertex set into two disjoint blocks, subject to a capacity bound, while minimizing the weight of nets that cross between them[5,18]. A graph joins pairs of vertices; a hypergraph $\mathcal{H} = (V, \mathcal{E})$ permits a net $e \in \mathcal{E}$ to contain any number of vertices. We write $n = |V|$, $m = |\mathcal{E}|$, $r_e = |e|$ and $w_e \geq 0$ for its weight. We first ask whether a binary energy preserves this partitioning objective.

Binary $x_i \in \{0, 1\}$ records the block of vertex i. For a graph with edge set E_2, the weighted edge-cut cost is

$$E_{\mathrm{G}}(\mathbf{x}) = \sum_{\{i,j\} \in E_2} w_{ij}(x_i - x_j)^2. \tag{1}$$

Each edge contributes its weight when its endpoints occupy different blocks. For a hypergraph, $k_e = \sum_{i \in e} x_i$ counts the vertices of net e assigned to one block. The cut-net objective is

$$E_{\mathrm{cut}}(\mathbf{x}) = \sum_{e} w_e \mathbb{I}(0 < k_e < r_e). \tag{2}$$

A crossing net is charged once, irrespective of the number of separated vertex pairs. For example, a four-vertex net split 1:3 or 2:2 costs one net weight in either case; a pair surrogate instead counts three or four crossing pairs. For two-vertex nets, cut-net and graph cut coincide.

The unit-load two-way problem also imposes balance: exact balance has $N = \sum_i x_i = b = \lfloor n/2 \rfloor$, whereas an allowed imbalance band specifies integer bounds on N. All reported cuts are scored on the original nets and accompanied by their actual imbalance. An allowed imbalance is an upper bound; a more balanced output remains feasible at that setting.

Quadratic unconstrained binary optimization (QUBO) expresses an energy as $\mathbf{z}^{\mathsf{T}}\mathbf{Q}\mathbf{z} + c_0$, where $\mathbf{z}$ is binary[11,12]. Higher-order unconstrained binary optimization (HUBO) also permits products of more than two variables. The

native cut has the exact HUBO form[18]

$$E_{\mathrm{H}}(\mathbf{x}) = \sum_e w_e \left[1 - \prod_{i\in e} x_i - \prod_{i\in e}(1-x_i)\right] + \lambda(N-b)^2. \quad (3)$$

The products identify all-one and all-zero nets; the quadratic penalty enforces the exact-balance target at sufficiently large λ. Quadratization replaces higher-order terms with quadratic interactions and auxiliary variables[13]. Here two binary auxiliaries per net give an exact QUBO after auxiliary minimization (Methods). Larger nets can therefore be represented by higher-order interactions or by additional binary spins. Replacing binary variables with diagonal Pauli operators supplies the corresponding quantum cost Hamiltonian. Its encoding fixes the endpoint; the driver and temporary catalysts determine a path to it. We next ask which representations preserve the target and provide a useful response before the labels become discrete.

The first obstacle is objective distortion. We compare two quadratic clique surrogates that sum $w_e k_e(r_e - k_e)$ or divide that quantity by $r_e - 1$. Both count pairs separated within each net. Even normalizing the pairs cannot make every crossing configuration of an arbitrary net cost exactly one. The difference affects the optimum, not just the quality of a heuristic.

Exact enumeration gives concrete counterexamples (Fig. 1). Each has 12 vertices, ten mixed-size nets and 924 balanced assignments. In example A, the native optimum is 7, but every minimum of the unit-pair surrogate has native cut 8. In example B, the native optimum is 5; the normalized surrogate selects native cut 6, while the unit-pair surrogate has minimum-state cuts ranging from 8 to 10. These are illustrative counterexamples, not estimates of average surrogate performance. Exact HUBO and the auxiliary-minimized QUBO recover the native minima in both examples.

The exact HUBO preserves the objective without auxiliaries, but requires interactions up to the net size. To examine its continuous response, consider independent binary vertex labels with probabilities $p_i = \Pr(x_i = 1)$. The uncertain state has $p_i = 1/2$. Under independent Bernoulli marginals, the expected HUBO cost has off-diagonal vertex curvature $-w_e 2^{3-r_e}$ at $p_i = 1/2$. Thus, the structural response around the uncertain state decays exponentially with net size. This is a local property of that variational representation, not a proof that all HUBO solvers fail. Table 1 separates objective distortion, interaction order and auxiliary cost. The complete QUBO retains the exact endpoint using two auxiliaries per net, and permits a temporary quadratic structural response. Its exact-balance penalty also creates a coordinate barrier: every balanced, auxiliary-minimized binary partition is a one-bit local minimum when $\lambda > D_{\max}$, where $D_{\max}$ is the maximum weighted incidence degree, regardless of its cut. Faithfulness alone therefore cannot supply a useful search dynamic (Methods).

Incidence controls balanced variational response

The encoding barriers motivate a temporary structural response while the vertex labels remain uncertain. We keep the complete exact QUBO and change its intermediate path. A pair term $\gamma(t)C_{\mathrm{pair}}$ redistributes interactions within nets, while an auxiliary term favors polarized auxiliary states:

$$C_{\mathrm{pair}} = \sum_e w_e a_e \sum_{i<j\in e} z_i z_j (x_i - x_j)^2,$$
$$C_{\mathrm{aux}}(t) = -\sum_e \kappa_e(t)(u_e - v_e)^2. \quad (4)$$

Here u_e, v_e are the two binary auxiliaries for net e. The scales z_i depend on incidence degree and local degree contrast; a_e preserves total coupling mass when these scales change. Both catalysts vanish at the endpoints. Intermediate per-spin temperatures supply another variational control, with their offsets also vanishing before the final original-QUBO phase (Fig. 2). Consequently the final binary objective is unchanged, and a penalty $\lambda > D_{\max}$ guarantees exactly balanced global minima (Methods). This guarantee concerns the encoding, not the output of a finite heuristic.

To identify the response controlled by these interactions, we examine the reduced free energy after minimizing each four-state auxiliary block. For even n, at uniform temperature and vertex

Table 1 Cut-net encodings

Construction	Order	Binary variables	Property relevant to cut-net search
Unit-pair clique	2	n	Different objective; example A selects cut 8 instead of 7.
Normalized clique	2	n	Different objective; example B selects cut 6 instead of 5.
Exact HUBO	$\leq r_{\max}$	n	Exact cost; uniform-state response decays as 2^{3-r_e}.
Complete auxiliary QUBO	2	$n+2m$	Exact cost after auxiliary minimization; extra spins and couplings.

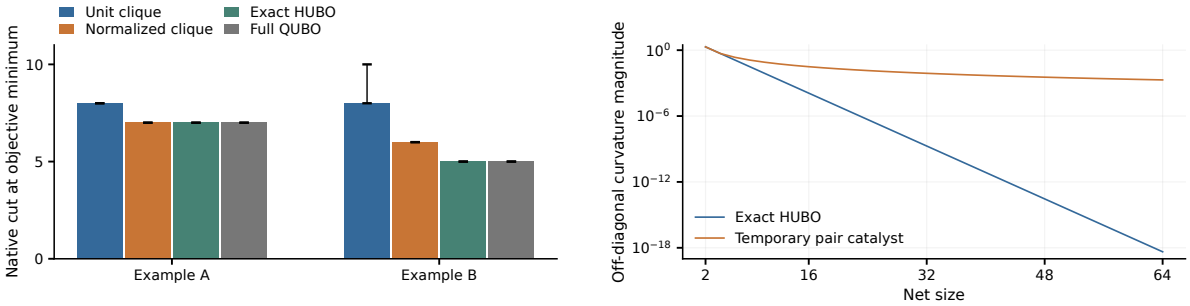


Figure 1 Objective fidelity and curvature; (a) Native cut among global minima of each objective on two exact counterexamples; Bar heights use the best native cut among tied minima; the whisker on example B's unit clique spans all native cuts among its minima, not an uncertainty interval; Exact HUBO and full QUBO recover the native optima; **(b)** Magnitudes of one net's off-diagonal curvature at independent $p_i = 1/2$: 2^{3-r} for the exact HUBO expectation, and $2/\max(1, \lfloor r^2/4 \rfloor)$ for a unit-strength normalized pair catalyst; The latter is temporary and is not substituted for the final cut objective

probabilities, the Hessian is

$$\begin{aligned}\mathbf{H} = {} & 4T\mathbf{I} + 2\lambda(\mathbf{1}\mathbf{1}^{\mathsf{T}} - \mathbf{I}) \\ & - \mathbf{B}^{\mathsf{T}} \operatorname{diag}(w_e^2 \chi_e / T)\mathbf{B} \\ & - 2\gamma \mathbf{A}_{\text{off}},\end{aligned} \tag{5}$$

where $\mathbf{B}$ is incidence, χ_e is the auxiliary susceptibility and $\mathbf{A}_{\text{off}}$ is the graph-dependent off-diagonal pair operator. Projection onto the zero-mean subspace isolates balanced fluctuations. Entropy stabilizes uncertainty; auxiliary susceptibility and catalytic geometry shift the unstable directions. Degree alone cannot determine that operator. The exact curvature expression agrees with automatic second derivatives on weighted test instances. Its minimum eigenvalue changes systematically with catalyst strength on the illustrated hypergraph (Fig. 2(c)).

Local incidence weights need not supply an extended direction across a large graph. We therefore also use zero-mean, unit-norm low-frequency modes $\mathbf{v}_\ell$ of incidence-derived Laplacians, with both unscaled and degree-normalized geometry. Spectral partitioning supplies the classical structural motivation[32,33]. The modes are computed from the original incidence. Their collective quadratic catalyst is

$$\begin{aligned} C_{\text{col}}(t) &= -\frac{\rho(t)}{2} \sum_\ell \theta_\ell \left[(\mathbf{v}_\ell^{\mathsf{T}} \mathbf{s})^2 - \|\mathbf{v}_\ell\|^2 \right], \\ s_i &= 2x_i - 1. \end{aligned} \tag{6}$$

with $\theta_\ell \geq 0$ and $\sum_\ell \theta_\ell = 1$. For one mode, its expected-energy Hessian adds $-4\rho(\mathbf{v}\mathbf{v}^{\mathsf{T}} - \operatorname{diag}(v_i^2))$. The mean-balance rank-one term annihilates $\mathbf{v}$, while this catalyst changes curvature along $\mathbf{v}$ by $-4\rho(1 - \sum_i v_i^4)$. Thus an extended mode supplies a strong balanced instability; a mode localized on one vertex supplies none. This statement concerns local variational curvature, not a quantum gap.

Vertices incident only to singleton nets have zero cut response, but still enter balance. We retain these spins and add a temporary quadratic coupling on a zero-sum random signed direction supported on them. This null-support catalyst encourages their balanced polarization without supplying a binary partition or repairing any vertex. Both collective coefficients vanish at the endpoints. A shared replica portfolio combines individual modes, mode subspaces, local catalysts and per-spin annealing. Component contrasts handle disconnected cut-active supports (Methods).

These calculations establish a local mechanism: the catalyst changes balanced variational curvature while preserving the final energy. The schematic overlap-gap and trapping panels in Fig. 2(d)–(f) illustrate the distinction from global solution geometry. An overlap gap does not require a flat surface or vanishing gradients, and neither its existence nor its removal follows from this Hessian. The next test is therefore empirical: whether the available directions produce useful partitions under a fixed search policy. A computed variational slice is provided in Supplementary Note 3. The same quadratic terms also define diagonal Ising operators in a transverse-field path (Methods), whose coherent dynamics require a separate test.

Solving catalytic QUBOs under capacity constraints

The local response analysis suggests search directions; their usefulness requires a solver test. We use a free-energy machine (FEM), a classical variational annealing method[31]. Each replica evolves vertex probabilities and four-state net-auxiliary distributions by balancing expected energy against entropy. Cooling produces binary candidates, which are evaluated on the original cut nets and capacity constraints (Methods). This separates the proposed energy path from the algorithm used to explore it.

KaHyPar supplies the classical reference. This multilevel hypergraph partitioner contracts vertices, constructs an initial partition and refines it as the contractions are undone[4,5]. Its partitions are evaluated independently and do not initialize FEM; Supplementary Note 1 describes the reference configurations.

We test FEM on 30 original hypergraphs with unit vertex loads and net weights at allowed imbalance 0%, 1% and 3%, with a separate 48-replica batch at each setting. One shared policy combines local, unscaled and degree-normalized groups with matched nearby random fields and 12,144 updates. Every vertex and net auxiliary remains in the QUBO; the bands add bounded slack (Methods). No classical partition initializes the search. These primary measurements use FP64. Table 2 pairs native cut with measured imbalance (%) at matching KaHyPar capacity settings. Bold marks the lower cut or charged time within each pair, including ties; the recorded references are not asserted optima.

The outcome depends strongly on capacity. FEM improves 15 of 30 recorded references at exact balance, but none at 1% or 3%; each band has four ties. Large gaps remain, including strict cuts of 2,544 against 1,204 on *IBM13* and 318 against 68 on *scircuit*. The expanded feasible set also does not guarantee a better finite search: *IBM01* changes from 325 at 0% to 335 at 3%, and *cfd1* from 2,064 to 2,250. The three settings are independent searches with equal trajectory counts, so strict results are not inserted into the band columns. These contrasts locate the current policy's usefulness at selected exact-balance settings, rather than establishing general competitiveness.

The exact-balance policy retains cuts 146 on *finan512*, 255 on *thermal1*, 624 on *G2_circuit* and 1,026 on *parabolic_fem*. KaHyPar's 3% runs on *finan512*, *G2_circuit* and *parabolic_fem* also return exactly balanced partitions, with cuts 146, 624 and 1,026. These remain feasible classical ties even when the separately configured 0% references are worse. Classical rounding of eight continuous incidence modes matches these three cuts and reaches 272 on *thermal1*; FEM improves that last cut by 17 nets. Spectral information alone explains several strong outcomes, and its rounded partitions are assessed separately.

The FEM score includes the specified original-QUBO one-bit descent, whose time is charged for all replicas. On 29 exact-balance instances, the winning raw partition is unchanged. On *scircuit*, the best feasible raw cut is 373; a different raw replica with count residual 2 reaches cut 318 through two vertex and four auxiliary flips. This exception is reported explicitly. Raw feasibility

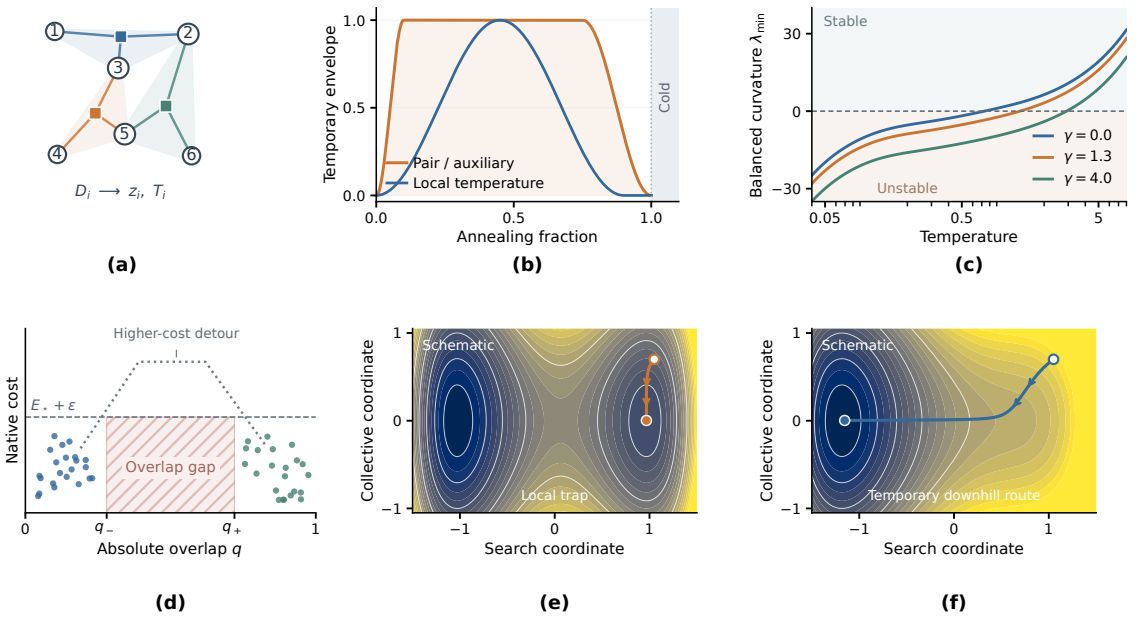


Figure 2 Catalyst landscapes; (a) A schematic shared-net hypergraph; squares represent nets and circles represent vertices, with larger circles indicating higher degree; Degree and local incidence contrast determine structural scales and temperature offsets; **(b)** Temporary pair/auxiliary envelopes and local-temperature offsets vanish before the cold phase; **(c)** Minimum balanced-subspace curvature from Equation (5) on example B, with penalty 0.01 and pair strengths 0, 1.3 and 4; Zero marks the local stability boundary; The curves are classical free-energy curvatures, not quantum spectral gaps; **(d)** Schematic OGP: near-optimal pairs have absolute overlap outside (q_-, q_+); the dotted detour leaves the near-optimal set; **(e)** An illustrative two-basin potential traps gradient descent in a local minimum; **(f)** A temporary term changes the descent direction and removes that local trap in the illustration; Contours indicate increasing potential; open circles mark the common initial state and filled circles the endpoints; Panels (d)–(f) are schematic, not measurements on benchmark graphs; The catalyst vanishes at the end of a search, restoring the original landscape; no OGP escape or measured barrier reduction is inferred

counts, cut differences and winner flip counts for all three balance settings are in Supplementary Note 4.

Twenty inputs informed earlier studies; ten extend the fixed policy to the full *IBM01–IBM18* family and three further matrix-derived hypergraphs. One base seed per graph does not establish seed robustness or fully unseen generalization. Group comparisons and mechanism controls are retained in Supplementary Notes 4 and 5.

A separate comparison asks whether useful catalytic paths also exist without net auxiliaries (Tables 3 and 4). On three seen inputs at exact balance, a native HUBO uses the shared vertex policy or a fixed mixture of four timing paths within the same 48-replica budget. Matched QUBO controls disable auxiliary polarization and share vertex fields, balance updates and global schedules. The off arms remove both couplings and local-temperature offsets, so the contrast measures their combined search policy. The mixture retains cut 146 on *finan512* and improves the shared HUBO cuts to 2,232 on *IBM13* and 116 on *rajat26*. Both latter cuts remain above KaHyPar. The corresponding QUBO mixture improves *IBM13* to 1,941 but regresses on *rajat26*, showing that binary endpoint equivalence does not make the variational paths interchangeable. These controls support selected HUBO outcomes; they do not replace the primary QUBO table or establish broad transfer.

Preprocessing and precision limit practical gains

Solution quality must be assessed together with the cost of obtaining it. We distinguish the primary pipeline measurements from kernel-fusion and precision controls, because their numerical paths and preprocessing differ.

Table 2 QUBO cut and measured imbalance (%);

	Allowed imbalance 0%		Allowed imbalance 1%		Allowed imbalance 3%	
Graph	FEM	KaHyPar	FEM	KaHyPar	FEM	KaHyPar
IBM01	325 (0.000)	**287** (0.000)	368 (0.988)	**220** (0.471)	335 (2.964)	**202** (2.760)
IBM02	477 (0.000)	**367** (0.000)	454 (0.990)	**350** (0.520)	420 (3.000)	**337** (2.989)
IBM03	**1,171** (0.000)	1,247 (0.000)	1,160 (0.994)	**962** (0.000)	1,143 (2.939)	**957** (0.199)
IBM04	**621** (0.000)	696 (0.000)	636 (0.996)	**598** (0.996)	609 (2.995)	**582** (1.898)
IBM05	1,901 (0.000)	**1,807** (0.000)	1,896 (0.402)	**1,725** (0.000)	1,895 (0.450)	**1,719** (1.036)
IBM06	**1,080** (0.000)	1,335 (0.000)	1,078 (0.560)	**1,023** (0.923)	1,080 (0.892)	**988** (1.994)
IBM07	**1,163** (0.000)	1,428 (0.000)	1,144 (0.997)	**948** (0.997)	1,206 (2.996)	**910** (2.992)
IBM08	1,238 (0.000)	**1,200** (0.000)	1,181 (0.998)	**1,140** (0.000)	1,211 (2.276)	**1,140** (0.000)
IBM09	756 (0.000)	**703** (0.000)	723 (0.996)	**627** (0.974)	685 (2.996)	**620** (2.663)
IBM10	2,070 (0.000)	**1,626** (0.000)	1,732 (0.746)	**1,340** (0.000)	1,720 (0.668)	**1,333** (0.349)
IBM11	**1,581** (0.000)	1,671 (0.000)	1,458 (0.998)	**1,079** (0.989)	1,417 (2.999)	**1,062** (2.891)
IBM12	**2,443** (0.000)	3,138 (0.000)	2,457 (0.788)	**1,996** (0.979)	2,492 (2.895)	**1,936** (2.476)
IBM13	2,544 (0.000)	**1,204** (0.000)	2,508 (1.000)	**848** (0.404)	1,921 (3.000)	**848** (0.404)
IBM14	3,259 (0.000)	**2,273** (0.000)	3,088 (1.000)	**1,849** (0.000)	3,577 (3.000)	**1,849** (0.000)
IBM15	4,573 (0.000)	**4,554** (0.000)	4,721 (0.999)	**2,839** (0.722)	4,145 (0.586)	**2,741** (1.868)
IBM16	**2,907** (0.000)	4,044 (0.000)	3,770 (1.000)	**2,117** (0.832)	3,544 (2.746)	**1,969** (2.941)
IBM17	3,446 (0.000)	**3,390** (0.000)	3,366 (0.999)	**2,351** (0.965)	2,784 (3.000)	**2,310** (2.600)
IBM18	**2,197** (0.000)	2,772 (0.000)	2,231 (0.537)	**1,961** (0.000)	2,326 (2.989)	**1,915** (1.355)
2cubes_sphere	**4,125** (0.000)	6,892 (0.000)	3,991 (0.989)	**3,922** (0.993)	3,951 (2.999)	**3,884** (2.993)
G2_circuit	**624** (0.000)	1,144 (0.000)	**624** (0.779)	**624** (0.000)	**624** (0.779)	**624** (0.000)
PGPgiantcompo	463 (0.000)	**396** (0.000)	467 (0.449)	**352** (0.000)	457 (2.959)	**350** (0.000)
airfoil_2d	375 (0.000)	**372** (0.000)	373 (0.999)	**328** (0.000)	342 (2.997)	**324** (2.899)
bcsstk29	372 (0.000)	**360** (0.000)	**360** (0.172)	**360** (0.000)	**360** (0.172)	**360** (0.000)
ca-CondMat	4,473 (0.000)	**4,202** (0.000)	4,406 (0.994)	**3,918** (0.916)	4,215 (3.000)	**3,855** (3.000)
cfd1	**2,064** (0.000)	3,121 (0.000)	2,174 (0.999)	**1,564** (0.725)	2,250 (2.998)	**1,564** (0.725)
finan512	**146** (0.000)	169 (0.000)	**146** (0.000)	**146** (0.000)	**146** (0.000)	**146** (0.000)
parabolic_fem	**1,026** (0.000)	1,256 (0.000)	**1,026** (0.013)	**1,026** (0.000)	**1,026** (0.013)	**1,026** (0.000)
scircuit	318 (0.000)	**68** (0.000)	277 (0.999)	**46** (0.513)	46 (2.451)	**42** (2.273)
shallow_water2	**768** (0.000)	909 (0.000)	768 (0.764)	**640** (0.000)	768 (0.601)	**640** (0.000)
thermal1	**255** (0.000)	291 (0.000)	257 (0.102)	**239** (0.111)	257 (0.099)	**239** (0.114)

Table 3 HUBO cut and measured imbalance;

Graph	Off	Shared catalyst	Mixed catalyst	KaHyPar
finan512	7,445 (0.000)	**146** (0.000)	**146** (0.000)	169 (0.000)
IBM13	4,755 (0.000)	2,385 (0.000)	2,232 (0.000)	**1,204** (0.000)
rajat26	345 (0.000)	181 (0.000)	116 (0.000)	**32** (0.000)

Table 5 reports the cost of the primary FP64 policy. It charges both mode constructions, graph setup, compilation/capture, all 48 trajectories, checkpoints and all-replica descent. CPU construction costs are summed even when jobs overlap. On *parabolic_fem*, the original CPU modes cost 1466.81 s and the strict GPU loop 467.52 s; the preprocessing dominates its 1971.66 s charged total. On *scircuit*, both an unsuccessful CPU mode attempt and its certified A100 replacement are charged. The longest entries therefore measure the complete historical pipeline, including costly preprocessing, rather than GPU arithmetic alone.

A separate precision control on *IBM01* reuses certified mode arrays across six encoding/capacity combinations (Supplementary Note 6). FP32 reduces search from 19.73–21.77 s to 14.32–15.84 s, while charged totals fall from 27.01–28.67 s to 21.66–22.76 s. The smaller pipeline gain reflects costs outside the accelerated search. Precision also changes the winning trajectory: strict QUBO cut worsens from 325 to 438, whereas 3% QUBO

Table 4 HUBO charged time (s);

Graph	Off	Shared catalyst	Mixed catalyst	KaHyPar
finan512	**75.62**	112.44	113.77	128.59
IBM13	**88.97**	138.38	139.20	130.33
rajat26	**48.85**	93.05	93.48	128.54

Table 5 QUBO charged time (s);

	Allowed imbalance 0%		Allowed imbalance 1%		Allowed imbalance 3%	
Graph	FEM	KaHyPar	FEM	KaHyPar	FEM	KaHyPar
IBM01	**25.35**	128.68	**20.73**	64.32	**21.98**	64.67
IBM02	**31.42**	128.30	**27.52**	64.61	**29.46**	64.44
IBM03	**33.19**	128.61	**29.23**	64.15	**29.79**	64.51
IBM04	**34.91**	128.67	**31.20**	64.23	**33.00**	65.10
IBM05	**35.75**	128.50	**32.00**	64.26	**33.43**	64.30
IBM06	**37.31**	129.22	**34.35**	64.66	**35.32**	66.49
IBM07	**49.79**	128.73	**48.05**	64.40	**49.88**	65.25
IBM08	**105.68**	128.37	103.51	**64.82**	105.71	**64.52**
IBM09	**59.74**	128.67	**58.02**	64.48	**59.28**	64.66
IBM10	**84.20**	129.23	82.91	**64.43**	85.31	**64.85**
IBM11	**78.65**	128.55	77.93	**64.41**	80.82	**64.90**
IBM12	**91.24**	128.81	90.88	**66.19**	91.72	**66.35**
IBM13	**103.27**	130.33	103.87	**65.48**	106.56	**64.40**
IBM14	212.18	**128.97**	219.53	**66.70**	220.02	**65.71**
IBM15	208.97	**65.63**	214.57	**64.59**	216.81	**69.89**
IBM16	297.77	**136.65**	304.09	**68.51**	307.32	**72.02**
IBM17	215.52	**144.28**	221.60	**66.23**	223.75	**65.19**
IBM18	252.76	**130.88**	260.83	**67.02**	263.85	**67.20**
2cubes_sphere	132.94	**129.14**	134.23	**67.97**	**134.80**	155.80
G2_circuit	163.52	**129.09**	170.21	**64.91**	173.42	**67.27**
PGPgiantcompo	**24.87**	128.57	**20.57**	64.31	**21.61**	64.40
airfoil_2d	**28.33**	128.35	**24.24**	64.16	**24.81**	64.23
bcsstk29	**32.47**	129.15	**28.11**	64.27	**29.45**	64.41
ca-CondMat	**40.99**	128.85	**36.71**	65.58	**37.52**	66.07
cfd1	**94.29**	130.68	93.73	**64.89**	95.12	**64.73**
finan512	**86.46**	128.59	86.11	**64.52**	88.30	**64.52**
parabolic_fem	1971.66	**133.58**	2004.51	**67.59**	2007.91	**80.87**
scircuit	3352.19	**132.23**	3419.80	**65.11**	3744.58	**66.07**
shallow_water2	**74.41**	128.99	73.75	**64.73**	75.30	**64.85**
thermal1	**106.93**	128.47	105.68	**64.90**	107.03	**66.02**

improves from 314 to 284. The cut changes have opposite signs, so these six runs on one input do not establish that FP32 generally improves or worsens quality. The measurements were made in separate runs; a broader paired comparison is required to assess the reproducibility of quality and timing changes.

This separates two practical limitations. Mode construction can dominate the historical pipeline, while repeated count-root and sparse actions still cost time inside the search. Component profiling and kernel-fusion controls identify both targets (Supplementary Note 6); they are separate protocols, not revised primary timings. Hardware and reference budgets also differ, so the recorded costs do not establish an equal-budget or equal-hardware speedup.

Coherent dynamics change target-partition probability

The classical FEM tests variational response at scale. To assess the local catalyst family's effect on quantum dynamics separately, we simulate eight six-vertex, 14-qubit complete encodings. This test does not include the large-instance collective catalyst.

At duration 12, mean probability of an optimal balanced vertex partition rises from 23.02% to 37.04%, improving on all eight cases. Full-QUBO ground-state probability rises from 2.31% to 3.12% on average, with one regression. Mean balance probability falls from 85.77% to 83.73% (Fig. 3). The catalyst improves the target-partition observable while slightly reducing feasibility and producing less consistent encoded-state success. The partition and auxiliary-state observables therefore answer different questions. Strict KaHyPar reaches the exact cut on all eight inputs in 2–4 ms. These classical simulations of quantum evolution demonstrate a catalyst effect, not quantum advantage.

Cut nets quantify communication

The final test asks what a cut reduction means for a shared-dependency workload. We construct a two-worker synchronous dataflow workload on *IBM01*, *IBM06* and *IBM15* connectivity. Each vertex has a numerical state; each net reduces its incident states and distributes that value to its local operators. Crossing nets exchange contributions between workers. Net payloads contain 8 or 32 double-precision values. The connectivity is from real circuit netlists, while operator functions are defined by the benchmark. These files lack signal directions and gate functions, so the workload is not gate-level circuit simulation.

For payload width d, the transmitted volume per step is exactly $V = 2d\, s_{\mathrm{fp64}} E_{\mathrm{cut}}$, with $s_{\mathrm{fp64}} = 8$ bytes. Instrumented two-process messages agree with this count in all 54 trials. All parallel outputs agree with the serial computation to within 1.12×10^{-16} in maximum absolute error. This establishes an application-level meaning for the cut-net objective: each crossing shared dependency contributes one fixed bidirectional payload, irrespective of its number of pins.

The application retains both base-seed partitions from the fixed portfolio and the strict KaHyPar partition. Averaging the two FEM seed volumes gives ratios to KaHyPar of 0.9948 on *IBM01*, 1.0210 on *IBM06* and 1.0890 on *IBM15*. The best *IBM15* seed matches KaHyPar's 4,554 crossing nets; the other has 5,365. No application result is selected after timing. At width 32, measured per-step medians are 5.63–5.84 ms for the *IBM01* FEM partitions versus 5.74 ms for KaHyPar, and 110.35–112.80 ms for *IBM15* versus 121.77 ms (Fig. 4). Three repeats per partition and payload are retained. These are observed local CPU microbenchmark times; communication volume, sparse-work distribution and process overhead jointly affect latency, so a cut reduction alone does not guarantee application acceleration.

Discussion

The central result is a connection between objective fidelity and controllable search response. Clique substitution can change the desired optimum; exact encoding removes that error but does not remove the variational and coordinate barriers. Incidence-dependent temporary interactions supply local and extended balanced directions while the final cut-net energy is restored. The Hessian explains this mechanism within the stated trial distribution, and the coherent tests show that its consequences depend on which success observable is measured.

The practical evidence is narrower than the mechanism. The shared policy improves selected exact-balance references but loses its comparative gains when capacity is relaxed. Spectral rounding and exactly balanced KaHyPar outputs already match the strong cuts on *finan512*, *G2_circuit* and *parabolic_fem*; the collective modes therefore carry useful classical information that must be included in any interpretation of the gains. Large remaining quality gaps and costly preprocessing limit the present solver. The single-input precision control raises a numerical question; it does not establish general FP32 degradation. FP32 addresses part of the computation, but broader paired quality and cost measurements are needed before revising benchmark claims.

The dataflow experiment supplies a direct reason to retain the cut-net objective: every crossing dependency adds a fixed exchanged payload. Its latency results also expose the gap between communication volume and application speed.

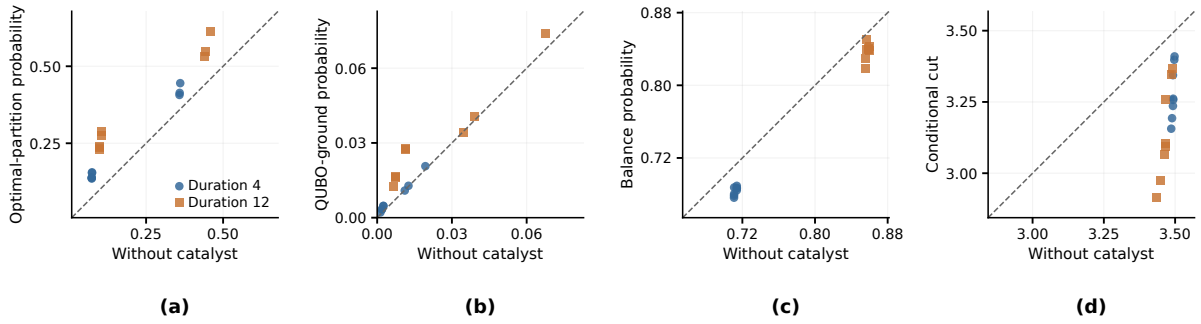


Figure 3 Coherent annealing; Eight matched 14-qubit encodings are simulated at durations 4 and 12; Horizontal axes show uncatalyzed observables and vertical axes show catalyzed observables; dashed lines indicate equality; **(a)** Optimal balanced vertex-partition probability, summed over auxiliary bits; **(b)** Full-QUBO ground-state probability; **(c)** Balance probability; **(d)** Native cut conditional on balance; Higher probability is preferred in (a)–(c) and lower cut in (d); Exact enumeration and strict KaHyPar agree on cuts 2 or 3 on every instance

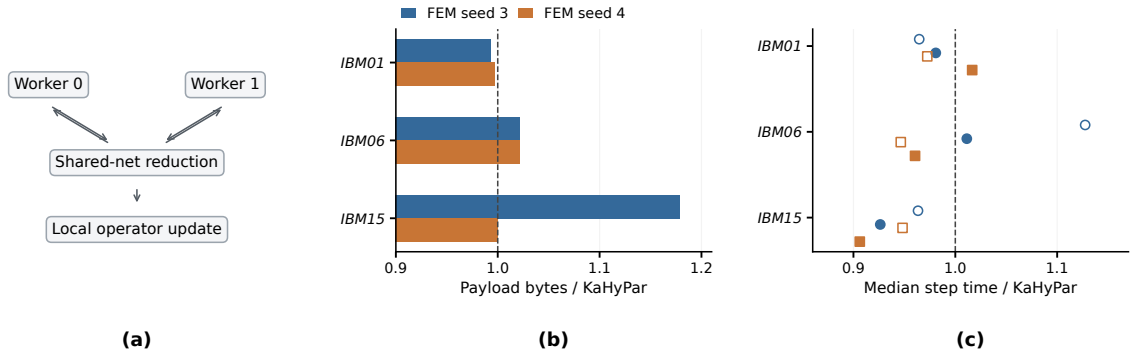


Figure 4 Circuit dataflow; (a) Shared-net reduction/broadcast and local numerical updates on two workers; **(b)** Measured transmitted-payload ratios to the strict KaHyPar partition at width 32; Both fixed FEM base seeds are shown; **(c)** Ratios of per-step median runtime from three repeats; Circles mark seed 3 and squares seed 4; open markers use width 8 and filled markers width 32; Values below one improve the corresponding KaHyPar application metric; The experiment measures two local CPU workers with synthetic operator functions on real circuit connectivity; Full cut, partitioning-time and application-time tables are given in Supplementary Note 7

Weighted operator costs, more processors and an actual simulator are needed to assess deployment. Likewise, scalable classical FEM and small coherent simulations provide complementary evidence; their combination does not establish a quantum advantage.

An overlap-gap claim would require a specified ensemble, threshold, algorithm class and scaling analysis. Local curvature and a few returned partitions cannot determine forbidden overlaps throughout the low-cost set. Correlated variational models, including cluster mean-field[34], autoregressive distributions[35] and tensor networks[36], could test which barriers arise from independent vertex marginals. Transverse catalysts require separate quantum experiments. The current framework makes these questions testable by keeping the objective, intermediate response and endpoint evaluation explicit.

Methods

We derive the exact cut-net encoding, balanced variational update and temporary incidence-based interactions. We then describe the partitioning benchmarks, coherent simulations and communication experiments used to evaluate their effects on search and computational cost.

Problem definition and exact endpoint

For nonnegative net weights, the objective is $E_{\text{cut}}(\mathbf{x}) = \sum_e w_e \mathbb{I}(0 < k_e < r_e)$ with $\sum_i x_i = b$. For odd n, block sizes are b and $n - b$. Two auxiliaries per nonempty net give

$$\min_{u_e, v_e \in \{0,1\}} \big[1 + (r_e - 1 - k_e)u_e + (k_e - 1)v_e\big] = \mathbb{I}(0 < k_e < r_e). \tag{7}$$

The complete QUBO multiplies each net term by w_e and adds the balance penalty.

Let $D_i = \sum_{e \ni i} w_e$ and $D_{\max} = \max_i D_i$ denote weighted incidence degrees. Equation (7) follows by examining the three cases $k_e = 0$, $0 < k_e < r_e$ and $k_e = r_e$. Minimizing the two auxiliary bits gives costs 0, 1 and 0, respectively. A state with imbalance $d = |\sum_i x_i - b|$ can be balanced by flipping d vertices. Its native cut can increase by at most $dD_{\max}$, whereas removing its penalty saves λd^2. Thus $\lambda > D_{\max}$ excludes imbalanced global minima. We use $D_{\max} + 1$ and retain the complete $n + 2m$ binary state for independent scoring.

This equality concerns binary energies after auxiliary minimization; it does not equate the HUBO and QUBO variational landscapes. At a fixed vertex distribution, the expectation of the auxiliary minimum can be smaller than the minimum expected energy of auxiliaries independent of vertices. Conditional auxiliaries that select a minimizing state for each vertex assignment recover the HUBO expectation. Our factorized vertex distribution and within-net auxiliary factors do not impose these vertex–auxiliary correlations. Auxiliary entropy and a quantum driver also affect thermodynamics and dynamics beyond binary endpoint equivalence.

For an allowed imbalance ϵ, let $c = \lceil n/2 \rceil$, $U = \lfloor (1 + \epsilon)c \rfloor$, $L = n - U$ and $N = \sum_i x_i$. A bounded binary slack $S = \sum_j a_j b_j$ spans all integers from 0 to $U - L$. Its weights are powers of two followed by the remaining range. The matched 1% and 3% QUBOs replace exact balance by

$$E_{\text{bal}} = \lambda(N - L - S)^2. \tag{8}$$

Minimizing slack at fixed vertices gives $\lambda \operatorname{dist}(N, [L, U])^2$. The same vertex-flip bound proves that $\lambda > D_{\max}$ excludes infeasible global minima. This uses $\lceil \log_2(U - L + 1) \rceil$ extra bits, while retaining every vertex and net auxiliary. No binary vertex is projected or repaired by a balance rule. Reported imbalance is $100[\max(n_0, n_1)/\lceil n/2 \rceil - 1]$; the smallest possible odd split therefore has zero reported imbalance.

The same coefficient implies an endpoint coordinate barrier. At a balanced state, flipping vertex i while fixing auxiliaries changes energy by

$$\Delta_i E = \lambda + (1 - 2x_i) \sum_{e \ni i} w_e(v_e - u_e) \geq \lambda - D_i > 0. \tag{9}$$

If auxiliaries are conditionally minimized, none of their single flips lowers energy either. Every such balanced state is therefore a one-bit local minimum, including poor cuts. One-bit descent can restore feasibility and auxiliary consistency but cannot improve these already feasible endpoints. This lemma is independently checked by exhaustive weighted small-instance enumeration and on 384 saved *finan512/thermal1* states. It concerns coordinate dynamics of the encoding, not OGP or a limitation on correlated continuous or quantum dynamics. The band encoding has the same one-bit barrier when $N = L + S$: vertex flips pay at least $\lambda - D_i$, and a single slack flip pays λa_j^2. Allowing more feasible block sizes does not remove this encoding-induced barrier.

Objective-fidelity checks

For clique objectives, we evaluate both unit pair weights and weights $1/\max(r_e - 1, 1)$. Exact enumeration scores every balanced assignment on 12 vertices. Fixed generation seeds 200–1199 define ten mixed-size nets; the first strict counterexample for each surrogate is retained, giving seeds 200 and 207 after eight examined graphs. All surrogate minimizers are included and the best native cut among ties is reported. These are explicitly selected mechanism examples, not a representative benchmark. All 924 assignments per example agree with the independently scored auxiliary QUBO after conditional auxiliary minimization. KaHyPar uses one ordinary seed-0 strict-balance run per example.

Variational distribution and update

The binary endpoint does not determine a unique continuous search. We therefore specify the trial distribution before defining its dynamics.

Mean-field annealing replaces discrete states by continuous marginals[37]. FEM follows this variational approach[31], with entropy and expected energy defined for an explicit trial distribution[38]. For replica a, the distribution is $q_a(x,u,v) = \prod_i \mathrm{Bern}(x_i;p_i^a)\prod_e q_e^a(u_e,v_e)$. The free-energy objective contains the expected QUBO and catalysts, vertex negative entropy $\sum_i T_i(t)[p_i\ln p_i+(1-p_i)\ln(1-p_i)]$, and auxiliary negative entropy $T(t)\sum_{e,u,v} q_e(u,v)\ln q_e(u,v)$. The four-state auxiliary block is minimized exactly at each vertex update. It retains within-edge auxiliary correlation but no vertex–vertex correlation. For $\bar{k}_e = \sum_{i\in e} p_i$, its nonconstant state energies are 0, $w_e(r_e-1-\bar{k}_e)-\kappa_e$, $w_e(\bar{k}_e-1)-\kappa_e$, and $w_e(r_e-2)$. The reduced block contribution is $w_e - T\ln\sum_{u,v}\exp[-E_e^{\mathrm{nc}}(u,v)/T]$, where E_e^{nc} denotes these four nonconstant energies. Supplementary Note 3 specifies the projected variational slice and the illustrative landscapes used in Fig. 2; the latter are schematic potentials, rather than measured hypergraph landscapes.

The expected balance cost is

$$\mathbb{E}\big[(\textstyle\sum_i x_i - b)^2\big] = (\textstyle\sum_i p_i - b)^2 + \sum_i p_i(1-p_i). \tag{10}$$

Dropping the variance term would change the variational objective. For the 1% and 3% runs, one correlated slack block is independent of the vertex and net blocks. If g_s counts the binary representations of slack value s, conditional minimization gives $q_s \propto g_s\exp[-\lambda_t(\mu-L-s)^2/T]$, where $\mu=\sum_i p_i$. Multiplicity retains the entropy of the underlying binary states. The expected balance term is $\lambda_t[(\mu-L-\mathbb{E}S)^2+\sum_i p_i(1-p_i)+\mathrm{Var}(S)]$. The current slack expectation provides the per-replica target in the same implicit vertex update, and is refreshed before each update. Binary readout minimizes slack and net auxiliaries with vertices fixed.

For the direct HUBO comparison, the trial distribution has only vertex Bernoulli factors and no auxiliary entropy. Its free energy is

$$F_{\mathrm{H}}(\mathbf{p},t) = \mathbb{E}(E_{\mathrm{cut}}+C_t) + \lambda_t\mathbb{E}[(N-b)^2] + \sum_i T_i(t)\phi(p_i). \tag{11}$$

Here $\phi(p) = p\ln p + (1-p)\ln(1-p)$ and the cut expectation uses the native HUBO products given above. The native cut gradient is $g_i = \sum_{e\ni i} w_e[\prod_{j\in e\setminus\{i\}}(1-p_j) - \prod_{j\in e\setminus\{i\}} p_j]$. Incidence products of log probabilities evaluate these terms stably; singleton contributions cancel exactly. Temporary pair and collective terms retain their quadratic expectations. The controlled QUBO arm uses the same vertex catalysts with auxiliary polarization set to zero; its conditional auxiliary entropy remains present.

FEM represents vertex marginals by sigmoid fields and separates sparse interaction gradients from the global count penalty. The implicit count correction is a scalar root with strictly positive derivative, so it can be solved with a bracketed update. This remains a soft-penalty variational step, without projection onto a balanced binary set. The field equations, preconditioning, clipping and numerical checks are specified in Supplementary Note 4.

Constructing temporary search paths

With the distribution fixed, local interactions and collective directions control the intermediate response. Their envelopes return every temporary term to zero before the original-objective readout.

Local incidence response

For the pair catalyst, the base coefficient is $a_e^{(0)} = 1/\max(1,\lfloor r_e^2/4\rfloor)$. Node scales take the form $z_i = (1+D_i)^{-\alpha}\exp(\xi s_i)$, where s_i is a bounded structural signal. Coefficients are rescaled so that $\sum_e w_e a_e \sum_{i<j\in e} z_i z_j$ equals its base value. Degree signals use $\ln(1+D_i)$; local contrast subtracts the weighted neighbor mean, counting repeated net co-occurrences. Signals are centered, standardized, clipped to two standard deviations, recentered and bounded; isolated vertices receive zero signal. Incidence operations compute these quantities without constructing a clique adjacency matrix.

Collective balanced directions

Let $\mathbf{B}$ be net-by-vertex incidence, $\mathbf{r}$ the net sizes and $a_e = 1/\max(1,\lfloor r_e^2/4\rfloor)$. The implicit operator is $\mathbf{L} = \mathrm{diag}(\mathbf{B}^{\mathsf{T}}(\mathbf{a}\odot\mathbf{r})) - \mathbf{B}^{\mathsf{T}}\mathrm{diag}(\mathbf{a})\mathbf{B}$. Its products use sparse incidence without storing clique edges. Let $\mathbf{D}_c = \mathrm{diag}((L_{ii})_i)$ on the cut-active support, and let $\mathbf{P}_c$ remove the mean separately within each connected component. We

also compute modes of

$$\mathbf{L}_{\mathrm{norm}} = \mathbf{P}_c \mathbf{D}_c^{-1/2} \mathbf{L} \mathbf{D}_c^{-1/2} \mathbf{P}_c. \tag{12}$$

These degree-normalized physical directions are not eigenvectors of the unscaled operator. Inactive vertices are excluded only from these eigenproblems and retained in the complete QUBO.

Each geometry supplies four zero-mean, orthonormal physical directions. On connected support we discard the constant mode. On disconnected support, random component contrasts occupy its nonconstant nullspace. If the largest component exceeds half the total vertex count, we keep one contrast and three internal directions; otherwise, up to four contrasts are retained. Internal modes use a shift on the component nullspace. The Lanczos solver uses tolerance 10^{-6}, maximum 20,000 iterations, subspace size 32 and probe seed 314159; absolute residuals are checked below 10^{-5}. Modes are padded with zeros on inactive spins.

For $\mathbf{m} = 2\mathbf{p} - \mathbf{1}$, one catalyst's exact Bernoulli expectation is $-\rho[(\mathbf{v}^\mathsf{T}\mathbf{m})^2 - \sum_i v_i^2 m_i^2]/2$, with gradient $-2\rho[\mathbf{v}(\mathbf{v}^\mathsf{T}\mathbf{m}) - \mathbf{v}^{\odot 2} \odot \mathbf{m}]$. The diagonal is removed because binary $s_i^2 = 1$. Weighted subspaces sum these quadratic expectations. The 48-replica policy divides equally among local, unscaled-mode and normalized-mode groups. All groups receive the same nearby random initial fields, with local, auxiliary and per-spin controls acting simultaneously. Supplementary Note 4 specifies the mode weights, strength grid and component-nullspace schedules.

For $f > 1$ inactive spins, each null direction assigns $1/\lfloor f/2 \rfloor$ to one random subset and $-1/\lceil f/2 \rceil$ to its complement, then normalizes. Thus it has zero mean for odd or even support sizes; one inactive spin cannot supply such a direction. *thermal1* has 248 inactive spins and *ca-CondMat* has 610. Fixed replica-dependent seeds match the two collective groups. Local replicas have no null coupling.

Schedules and replica allocation

Temporary local temperatures are $T_i^a(t) = T(t) \exp[\omega(t)\zeta_a s_i]$, with $\omega(t) = \sin^2(\pi t/0.9)$ for normalized annealing time $0 \le t < 0.9$ and zero thereafter. The pair and auxiliary envelopes rise over the first 10% of annealing and decay over the final 25%, using squared-sine ramps. Auxiliary strengths multiply $w_e \max(r_e/2 - 1, 0)$. The balance penalty remains 0.01 for the first 90% and increases geometrically to $D_{\max} + 1$ over the final 10%. In the primary benchmark, temperature decreases exponentially from 8 to 0.02 over 12,000 updates; 144 additional updates retain the final temperature, full penalty and zero catalysts. Fixed checkpoints occur every 128 updates and at both phase boundaries. Selection uses the original full-QUBO energy.

Each 16-replica batch starts from a shared normal field with standard deviation 0.01 plus independent perturbations with standard deviation 0.001. Base seeds determine both fields reproducibly. The fixed portfolio combines negative and positive contrast weighting, degree-temperature biases, local-contrast cooling, inverse-degree geometry, a geometry transition and uniform auxiliary alternatives. Supplementary Note 4 gives the profile parameters. Replicas evolve as independent variational starts, without the temperature-swap transitions of exchange Monte Carlo[39]. Per-spin temperature offsets are classical variational controls; they do not implement the inhomogeneous transverse-field schedules studied in quantum annealing[40]. No graph name determines a profile or parameter.

The collective envelope rises over annealing fractions 0–0.15 and falls over 0.80–1; the null envelope rises over 0.55–0.70 and falls over 0.96–1. Squared-sine ramps give zero endpoints. Null strengths multiply $D_{\max} + 1$ by $\{0.5, 1, 2, 4\}$.

Temperature scheduling follows classical annealing[41]. Separate classical references stably sort each of the eight modes and assign exactly $\lfloor n/2 \rfloor$ vertices to one block. These binary partitions never enter FEM.

Local response and mechanism checks

The reduced free energy connects the specified path to a testable local response. We evaluate its balanced-subspace curvature, then check the analytic expressions against independent derivatives and enumeration.

For even n, uniform vertex probabilities are a stationary point of the reduced block-variational objective when the catalyst has no linear field.

At uniform temperature, Eq. (5) gives its vertex-probability Hessian, where $\mathbf{A}_{\text{off}}$ contains the off-diagonal pair weights $\sum_{e\ni i,j} w_e a_e z_i z_j$, and

$$\chi_e = \left[1 + \exp(-\kappa_e/T)\cosh\big(w_e(r_e/2-1)/T\big)\right]^{-1}. \tag{13}$$

Indeed, $\chi_e = \mathrm{Var}_{q_e}(v_e - u_e)$ at the uniform state; minimizing the auxiliary Gibbs block contributes its negative susceptibility to the Hessian. The entropy Hessian is $4T\mathbf{I}$, and the exact balance expectation contributes $2\lambda(\mathbf{1}\mathbf{1}^{\mathsf{T}} - \mathbf{I})$. The relevant balanced-subspace operator is $\mathbf{PHP}$, with $\mathbf{P} = \mathbf{I} - \mathbf{1}\mathbf{1}^{\mathsf{T}}/n$. Per-spin temperatures replace the entropy term by $4\,\mathrm{diag}(T_i)$. This expression is checked against automatic second derivatives of the complete reduced free energy on weighted small hypergraphs, with and without auxiliary polarization. Matrix-free incidence products evaluate the operator without storing a clique matrix. Its eigenvalues are local classical free-energy curvatures, not the eigenvalue gaps of Eq. (14).

The exact HUBO Bernoulli expectation is $\sum_e w_e[1 - \prod_{i\in e} p_i - \prod_{i\in e}(1-p_i)]$. Differentiating twice at independent $p_i = 1/2$ gives zero diagonal curvature and off-diagonal $-w_e 2^{3-r_e}$ within a net. The pair catalyst has off-diagonal $-2\gamma w_e a_e z_i z_j$. For Fig. 1(b), weights/scales and strength equal one, and $a_e = 1/\max(1, \lfloor r_e^2/4 \rfloor)$. Fig. 2(c) uses example B, a penalty of 0.01 and 100 logarithmically spaced temperatures from 0.04 to 8. Auxiliary polarization is zero; exact balanced-subspace eigenvalues are computed from the verified matrix-free Hessian.

Scope of the geometry analysis

For a fixed cut threshold τ, define the feasible low-cost set $\mathcal{S}_\tau = \{\mathbf{x} : \sum_i x_i = b,\ E_{\text{cut}}(\mathbf{x}) \le \tau\}$. The complement-invariant vertex overlap is $q(\mathbf{x},\mathbf{y}) = |n^{-1}\sum_i (2x_i-1)(2y_i-1)|$. An overlap-gap statement requires a forbidden interval of such overlaps between appropriate low-cost states, with its ensemble and threshold specified. The present real-graph runs and six-vertex coherent examples do not establish that property. Nor does a catalyst automatically evade an algorithmic obstruction: the resulting algorithm may still belong to a class for which a stability-based theorem applies. The curvature analysis concerns local variational stability only.

Partition evaluation and computational cost

The partitioning study uses 30 circuit-netlist, matrix-derived and network hypergraphs from public benchmark collections[42,43], with two blocks and unit loads and net weights. At each allowed imbalance (0%, 1%, 3%), one 48-replica batch receives 12,000 annealing and 144 cold updates, giving 4,320 primary trajectories. The policy is fixed before evaluating each cohort. Initial fields are random and are shared across matched controls; reference and rounded spectral partitions never initialize FEM.

Every replica retains its lowest original-QUBO checkpoint energy at 128-update intervals and phase boundaries. Vertex signs give raw labels; net auxiliaries and slack are minimized with vertices fixed. Original- polynomial one-bit descent then runs on all 48 replicas, without paired moves, binary projection or a feasibility filter on proposed moves. Initial fields and complete raw and selected states are independently reconstructed and scored. Feasibility requires the exact integer capacity and zero auxiliary gap; failed or incomplete outputs are retained. Readout and validation rules are given in Supplementary Note 8.

Computational cost includes mode construction, numerical setup, the complete annealing batch and checkpoints, and descent on all replicas. The primary search uses double precision on a GPU; mode construction is performed on a CPU, with safeguarded updates and a certified mode-construction alternative for *scircuit* specified in Supplementary Notes 4 and 6. Both mode-construction costs are charged even when the jobs overlap.

KaHyPar receives the same original hypergraph, cut objective and integer capacity. Its returned partitions are independently rescored, and measured elapsed time accompanies each result. The ordinary and evolutionary reference configurations are described in Supplementary Note 1; hardware and timing definitions are given in Supplementary Note 6. The comparisons assess quality at matching capacity, while different hardware and time budgets limit speedup interpretations.

Direct-encoding, schedule-mixture, penalty-feedback and precision controls address distinct

numerical questions and retain their specified budgets (Supplementary Notes 5 and 6). They are evaluated separately from the primary quality and timing results. A single-input precision control cannot establish a general quality effect.

Coherent quantum experiment

The FEM search above is classical. The independent coherent test replaces its variational updates with explicit evolution under the encoded Ising Hamiltonian and measures several endpoint observables.

Hamiltonian path

Binary x_i is represented by $\widehat{\mathbf{x}}_i = (\mathbf{I} - \widehat{\mathbf{Z}}_i)/2$, where $\widehat{\mathbf{X}}_i$ and $\widehat{\mathbf{Z}}_i$ are Pauli operators; the QUBO and temporary terms become diagonal Ising operators. A transverse-field path is

$$\begin{aligned} \widehat{\mathbf{H}}(s) = &-(1-s)\sum_j \widehat{\mathbf{X}}_j \\ &+ s\widehat{\mathbf{E}}_{\mathrm{QUBO}} + f(s)\widehat{\mathbf{C}}, \\ f(0) = f(1) &= 0. \end{aligned} \tag{14}$$

Our diagonal catalysts differ from the transverse XX catalysts in Refs.[20,22]. Annealing-path choices can change perturbative crossings even with a fixed final problem[44]. They are temporary interactions, not a catalytic quantum state whose marginal must return unchanged. FEM does not simulate unitary quantum evolution. Dense balance couplings and collective couplings, together with encoding auxiliaries, impose quantum connectivity and precision costs that the sparse classical implementation does not remove.

Evolution and observables

Eight synthetic hypergraphs use six vertices and four hyperedges of sizes 2, 3, 4 and 5, with fixed generation seeds 20–27. They require 14 qubits under Eq. (7). We evolve the complete state vector from $|+\rangle^{\otimes 14}$ using double-precision complex arithmetic. We simulate $\widehat{\mathbf{H}}(s) = -(1-s)\sum_j \widehat{\mathbf{X}}_j + [s\widehat{\mathbf{E}}_{\mathrm{QUBO}} + 4s(1-s)\widehat{\mathbf{C}}]/\lambda$, with durations 4 and 12 in units $\hbar = 1$. The diagonal catalyst uses pair strength 1.3 and auxiliary strength 0.55. Uncatalyzed runs set its envelope to zero. Midpoint Strang splitting alternates half diagonal phases, exact commuting single-qubit $\widehat{\mathbf{X}}$ rotations, and half diagonal phases, with time step at most 0.05. The first two instances are repeated at 0.025 for convergence checks. Exact enumeration supplies the optimal balanced cut and full-QUBO ground subspace. We report their total probabilities, rather than selecting one measurement or inferring quantum runtime from classical simulation time.

Circuit communication workload

To give a returned cut an application interpretation, partitions are frozen before execution of a shared-dependency workload. Its protocol separates transmitted payload from measured latency.

The application freezes three graphs, both raw fixed-policy base seeds (3,4), the strict KaHyPar partition, widths 8 and 32, 16 measured steps, three warm-up steps and three repeats before outcomes. Trial order is randomized. The partitions are selected before the application trials and are not supplied back to the QUBO search. All runs start from the same graph/payload-specific numerical state.

Let $\mathbf{X}$ be vertex states, $\mathbf{B}$ incidence, $\mathbf{r}$ net sizes and $\mathbf{d}_{\mathrm{inc}} = (D_1, \ldots, D_n)^{\mathsf{T}}$ vertex incidence degrees. Each step evaluates

$$\begin{aligned} \mathbf{Y} &= \mathrm{diag}(\mathbf{r})^{-1}\mathbf{B}\mathbf{X}, \\ \mathbf{G} &= \mathrm{diag}(\max(\mathbf{d}_{\mathrm{inc}}, 1))^{-1}\mathbf{B}^{\mathsf{T}}\mathbf{Y}, \\ \mathbf{X}' &= \tanh(0.85\mathbf{X} + 0.15\mathbf{G} + \mathbf{Z}). \end{aligned} \tag{15}$$

The bias $\mathbf{Z}$ and initial states use a fixed seeded normal distribution. Each worker computes local net reductions. Only crossing-net vectors are sent to the other worker through a bidirectional process pipe; uncut nets require no payload. The two contributions produce the same net means as a serial reference. Actual sent/received buffer lengths verify $2d\times 8$ transmitted bytes per crossing net per step, excluding message headers. Received bytes equal transmitted bytes. Maximum absolute error to the serial calculation is recorded for every trial.

The experiment uses two local CPU processes, each pinned to one available logical CPU with numerical-library threads limited to one. Reported loop time is the larger worker elapsed time, excluding graph construction, process startup, warm-up and final-state collection. Saved cuts and original partitioning times

are reported alongside application runtimes. We report median times from three repeats; Supplementary Note 7 gives the ranges and results for both partition seeds.

Data availability

Aggregate benchmark results and input provenance are available at https://github.com/yao-baijian/qubo-hypergraph-partition/tree/main/docs/benchmarks.

Code availability

The reference FEM implementation and partitioning interface are available at https://github.com/yao-baijian/qubo-hypergraph-partition.

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